

Date: 2026-07-13
Request ID: 100048725

Microwave				
Analyte(s)	Ca, Cr, Cu, Fe, K, Na, Ni, Zn			
SOP#				
Acid Cocktail	9 mL HCl, 3 mL HNO3, 6 mL H2SO4			
Rack	7			
Vessel	PTFE			
Stir	90%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200128458_1	1.2525	50		
200128458_2	1.2525	50		
200128458_3	1.2543	50		
200128459_1	1.2555	50		
200128459_2	1.2550	50		
200128459_3	1.2544	50		
200128460_1	1.2531	50		
200128460_2	1.2526	50		
200128460_3	1.2577	50		
200128461_1	1.2557	50		
200128461_2	1.2545	50		
200128461_3	1.2561	50		

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128458_1	1/1	3.552	141.80	0.01
200128458_2	1/1	3.531	140.96	0.01
200128458_3	1/1	3.477	138.59	0.01

ca result: 140.45 ppm

ca result: 0.01 %

ca lot: Manufacturer: Assurance, 28-35CAY exp: 2026-10-30

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200128458_1	1/1	1.541	61.53	0.01
200128458_2	1/1	1.565	62.49	0.01
200128458_3	1/1	1.557	62.08	0.01

cr result: 62.03 ppm

cr result: 0.01 %

cr lot: Manufacturer: Ricca, 45120423 exp: 2027-05-31

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128458_1	1/1	3.153	125.87	0.01
200128458_2	1/1	3.207	128.02	0.01
200128458_3	1/1	3.184	126.92	0.01

cu result: 126.94 ppm

cu result: 0.01 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200128458_1	1/1	2.132	85.09	0.01
200128458_2	1/1	2.159	86.17	0.01
200128458_3	1/1	2.047	81.60	0.01

fe result: 84.29 ppm

fe result: 0.01 %

fe lot: Manufacturer: Assurance, 28-180FEY exp: 2027-07-30

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200128458_1	10/1	0.356	142.12	0.01
200128458_2	10/1	0.341	135.93	0.01
200128458_3	10/1	0.321	127.96	0.01

k result: 135.33 ppm

k result: 0.01 %

k lot: Manufacturer: Assurance, 28-6KY exp: 2026-08-30

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200128458_1	10/1	0.544	217.17	0.02
200128458_2	10/1	0.520	207.58	0.02
200128458_3	10/1	0.507	202.10	0.02

na result: 208.95 ppm

na result: 0.02 %

na lot: Manufacturer: Assurance, 27-159NAY exp: 2026-08-30

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200128458_1	1/1	2.026	80.88	0.01
200128458_2	1/1	2.055	82.04	0.01
200128458_3	1/1	1.993	79.45	0.01

ni result: 80.79 ppm

ni result: 0.01 %

ni lot: Manufacturer: Assurance, 28-130NIT exp: 2026-10-30

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200128458_1	1/1	1.373	54.81	0.01
200128458_2	1/1	1.390	55.49	0.01
200128458_3	1/1	1.342	53.48	0.01

zn result: 54.59 ppm

zn result: 0.01 %

zn lot: Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128459_1	1/1	3.277	130.51	0.01
200128459_2	1/1	3.188	127.03	0.01
200128459_3	1/1	3.315	132.13	0.01

ca result: 129.89 ppm

ca result: 0.01 %

ca lot: Manufacturer: Assurance, 28-35CAY exp: 2026-10-30

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200128459_1	1/1	1.566	62.38	0.01
200128459_2	1/1	1.510	60.15	0.01
200128459_3	1/1	1.572	62.65	0.01

cr result: 61.72 ppm

cr result: 0.01 %

cr lot: Manufacturer: Ricca, 45120423 exp: 2027-05-31

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128459_1	1/1	3.985	158.70	0.02
200128459_2	1/1	4.063	161.87	0.02
200128459_3	1/1	4.054	161.59	0.02

cu result: 160.72 ppm

cu result: 0.02 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200128459_1	1/1	1.821	72.52	0.01
200128459_2	1/1	1.710	68.11	0.01
200128459_3	1/1	1.773	70.67	0.01

fe result: 70.43 ppm

fe result: 0.01 %

fe lot: Manufacturer: Assurance, 28-180FEY exp: 2027-07-30

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200128459_1	10/1	0.362	144.17	0.01
200128459_2	10/1	0.312	124.30	0.01
200128459_3	10/1	0.357	142.30	0.01

k result: 136.92 ppm

k result: 0.01 %

k lot: Manufacturer: Assurance, 28-6KY exp: 2026-08-30

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200128459_1	10/1	0.473	188.17	0.02
200128459_2	10/1	0.449	178.69	0.02
200128459_3	10/1	0.479	190.93	0.02

na result: 185.93 ppm

na result: 0.02 %

na lot: Manufacturer: Assurance, 27-159NAY exp: 2026-08-30

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200128459_1	1/1	2.592	103.23	0.01
200128459_2	1/1	2.578	102.71	0.01
200128459_3	1/1	2.488	99.17	0.01

ni result: 101.70 ppm

ni result: 0.01 %

ni lot: Manufacturer: Assurance, 28-130NIT exp: 2026-10-30

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200128459_1	1/1	1.475	58.72	0.01
200128459_2	1/1	1.244	49.56	0.00
200128459_3	1/1	1.170	46.64	0.00

zn result: 51.64 ppm

zn result: 0.01 %

zn lot: Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128460_1	10/2	2.776	553.73	0.06
200128460_2	10/2	2.839	566.62	0.06
200128460_3	10/2	2.688	534.21	0.05

ca result: 551.52 ppm

ca result: 0.06 %

ca lot: Manufacturer: Assurance, 28-35CAY exp: 2026-10-30

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200128460_1	10/2	2.402	479.21	0.05
200128460_2	10/2	2.085	416.13	0.04
200128460_3	10/2	2.145	426.37	0.04

cr result: 440.57 ppm

cr result: 0.04 %

cr lot: Manufacturer: Ricca, 45120423 exp: 2027-05-31

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128460_1	10/2	2.706	539.93	0.05
200128460_2	10/2	2.666	532.16	0.05
200128460_3	10/2	2.619	520.59	0.05

cu result: 530.89 ppm

cu result: 0.05 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200128460_1	10/2	2.505	499.76	0.05
200128460_2	10/2	2.782	555.25	0.06
200128460_3	10/2	2.756	547.83	0.05

fe result: 534.28 ppm

fe result: 0.05 %

fe lot: Manufacturer: Assurance, 28-180FEY exp: 2027-07-30

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200128460_1	25/1	0.957	954.63	0.10
200128460_2	25/1	0.929	927.07	0.09
200128460_3	25/1	0.919	913.37	0.09

k result: 931.69 ppm

k result: 0.09 %

k lot: Manufacturer: Assurance, 28-6KY exp: 2026-08-30

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200128460_1	25/2	1.105	551.13	0.06
200128460_2	25/2	1.133	565.32	0.06
200128460_3	25/2	1.135	564.03	0.06

na result: 560.16 ppm

na result: 0.06 %

na lot: Manufacturer: Assurance, 27-159NAY exp: 2026-08-30

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200128460_1	10/2	2.397	478.21	0.05
200128460_2	10/2	2.854	569.62	0.06
200128460_3	10/2	2.801	556.77	0.06

ni result: 534.87 ppm

ni result: 0.05 %

ni lot: Manufacturer: Assurance, 28-130NIT exp: 2026-10-30

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200128460_1	10/2	2.167	432.33	0.04
200128460_2	10/2	2.810	560.83	0.06
200128460_3	10/2	2.772	551.01	0.06

zn result: 514.72 ppm

zn result: 0.05 %

zn lot: Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128461_1	10/2	2.597	517.04	0.05
200128461_2	10/2	2.611	520.33	0.05
200128461_3	10/2	2.875	572.21	0.06

ca result: 536.53 ppm

ca result: 0.05 %

ca lot: Manufacturer: Assurance, 28-35CAY exp: 2026-10-30

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200128461_1	10/2	2.222	442.38	0.04
200128461_2	10/2	2.447	487.64	0.05
200128461_3	10/2	2.157	429.30	0.04

cr result: 453.11 ppm

cr result: 0.05 %

cr lot: Manufacturer: Ricca, 45120423 exp: 2027-05-31

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128461_1	10/2	2.620	521.56	0.05
200128461_2	10/2	2.799	557.79	0.06
200128461_3	10/2	2.583	514.09	0.05

cu result: 531.15 ppm

cu result: 0.05 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200128461_2	10/2	2.926	583.10	0.06
200128461_3	10/2	2.470	491.60	0.05

fe result: 537.35 ppm

fe result: 0.05 %

fe lot: Manufacturer: Assurance, 28-180FEY exp: 2027-07-30

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200128461_1	25/1	0.887	882.97	0.09
200128461_2	25/1	0.910	906.74	0.09
200128461_3	25/1	0.908	903.59	0.09

k result: 897.77 ppm

k result: 0.09 %

k lot: Manufacturer: Assurance, 28-6KY exp: 2026-08-30

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200128461_1	25/2	1.091	543.02	0.05
200128461_2	25/2	1.087	541.55	0.05
200128461_3	25/2	1.087	540.86	0.05

na result: 541.81 ppm

na result: 0.05 %

na lot: Manufacturer: Assurance, 27-159NAY exp: 2026-08-30

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200128461_1	10/2	2.761	549.59	0.05
200128461_2	10/2	2.670	532.08	0.05
200128461_3	10/2	2.689	535.09	0.05

ni result: 538.92 ppm

ni result: 0.05 %

ni lot: Manufacturer: Assurance, 28-130NIT exp: 2026-10-30

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200128461_1	10/2	2.609	519.38	0.05
200128461_2	10/2	2.589	515.89	0.05
200128461_3	10/2	2.491	495.68	0.05

zn result: 510.32 ppm

zn result: 0.05 %

zn lot: Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$